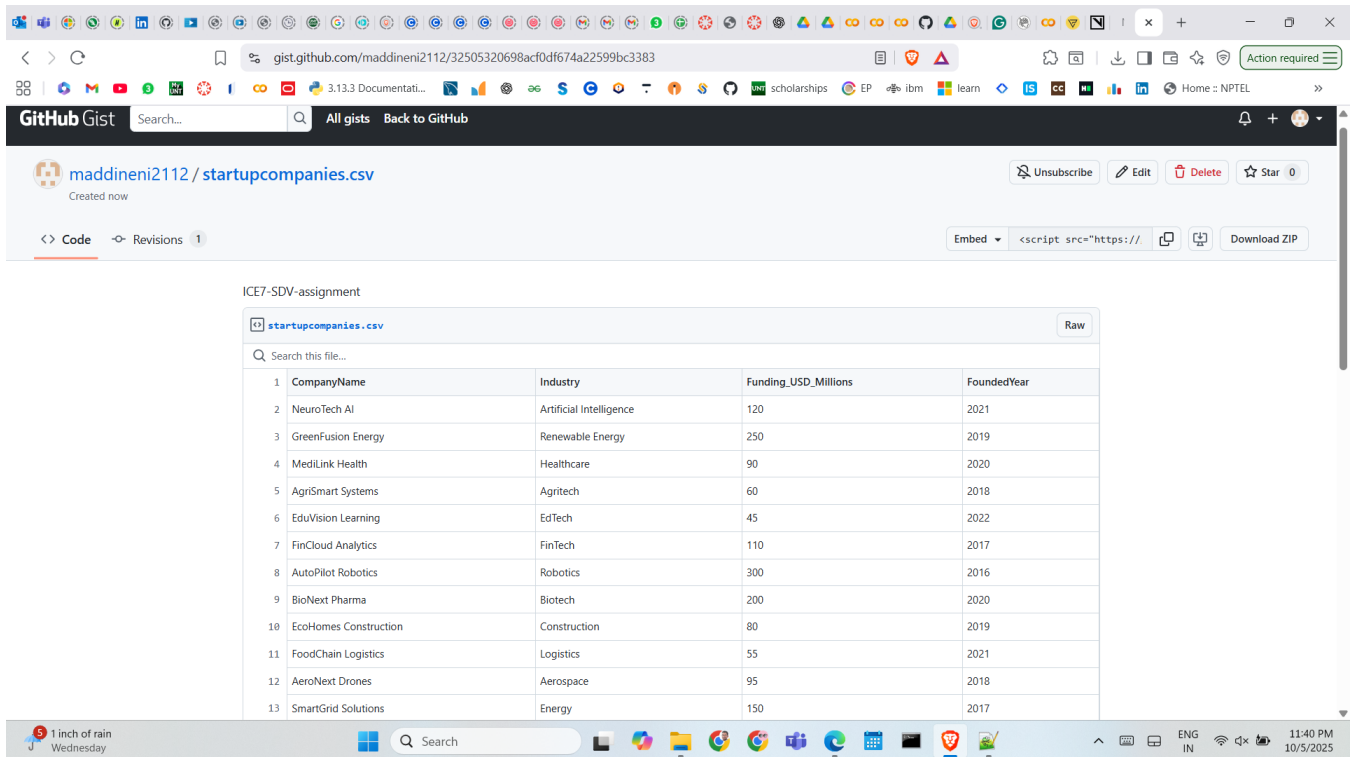


CSCE 5320: Scientific Data Visualization

Assignment 05

1. Create a csv dataset of your own or refer any data set and copy its contents
2. Then go to gist and add the csv file name, and its contents that we copied and create a public gist, then click on the raw and copy the .csv link

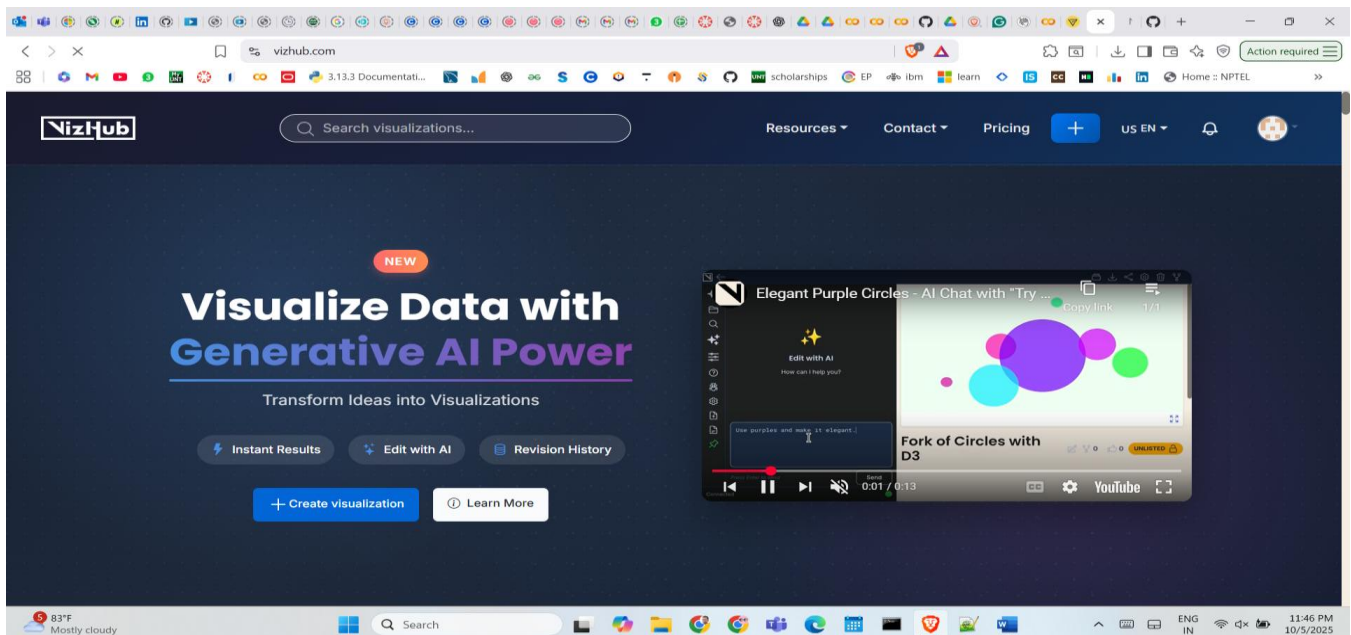


The screenshot shows a GitHub Gist page for a file named 'startupcompanies.csv'. The gist is created by 'maddineni2112' and is public. It contains a CSV dataset with 13 rows of startup information. The table has four columns: CompanyName, Industry, Funding_USD_Millions, and FoundedYear.

1	CompanyName	Industry	Funding_USD_Millions	FoundedYear
2	NeuroTech AI	Artificial Intelligence	120	2021
3	GreenFusion Energy	Renewable Energy	250	2019
4	MediLink Health	Healthcare	90	2020
5	AgriSmart Systems	AgriTech	60	2018
6	EduVision Learning	EdTech	45	2022
7	FinCloud Analytics	FinTech	110	2017
8	AutoPilot Robotics	Robotics	300	2016
9	BioNext Pharma	Biotech	200	2020
10	EcoHomes Construction	Construction	80	2019
11	FoodChain Logistics	Logistics	55	2021
12	AeroNext Drones	Aerospace	95	2018
13	SmartGrid Solutions	Energy	150	2017

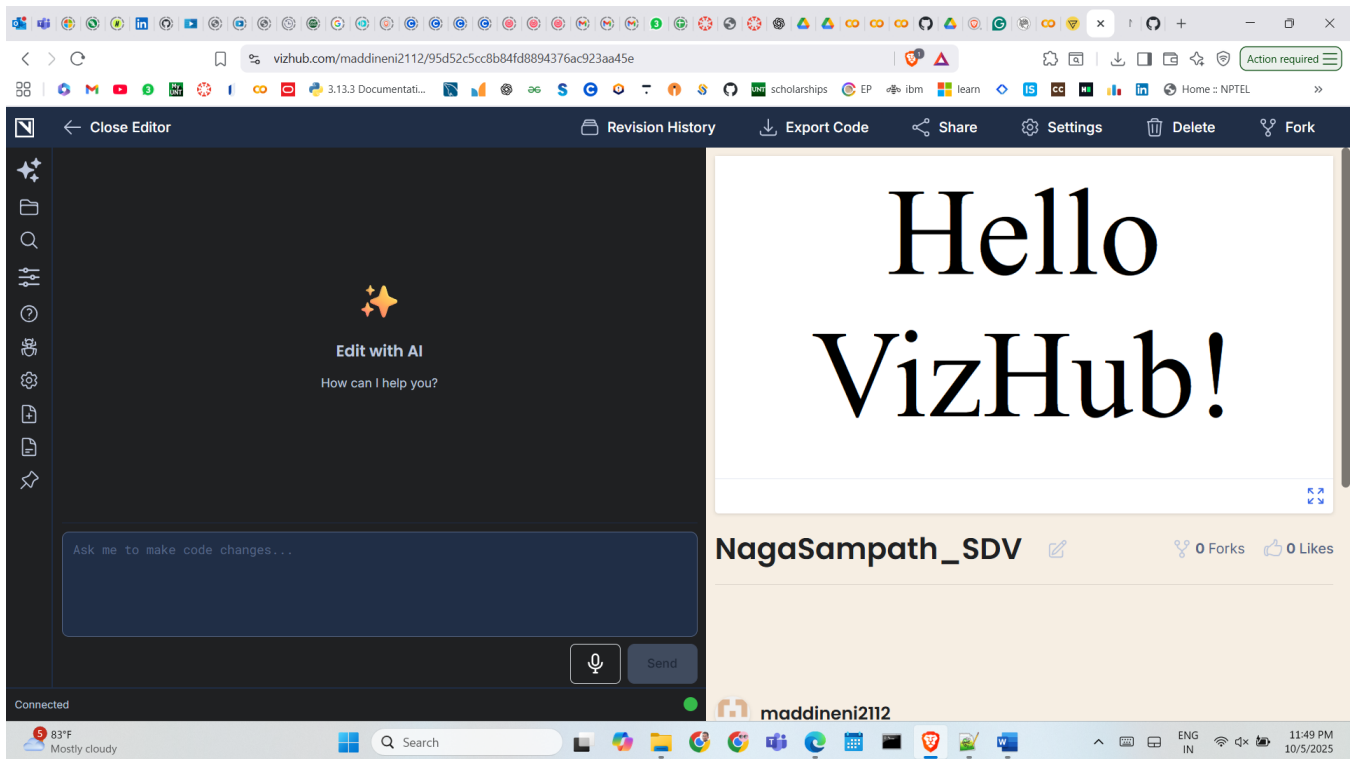
Gist dataset link: <https://gist.github.com/maddineni2112/32505320698acf0df674a22599bc3383>

3. Next to viz hub and create an account using git and create visualization.

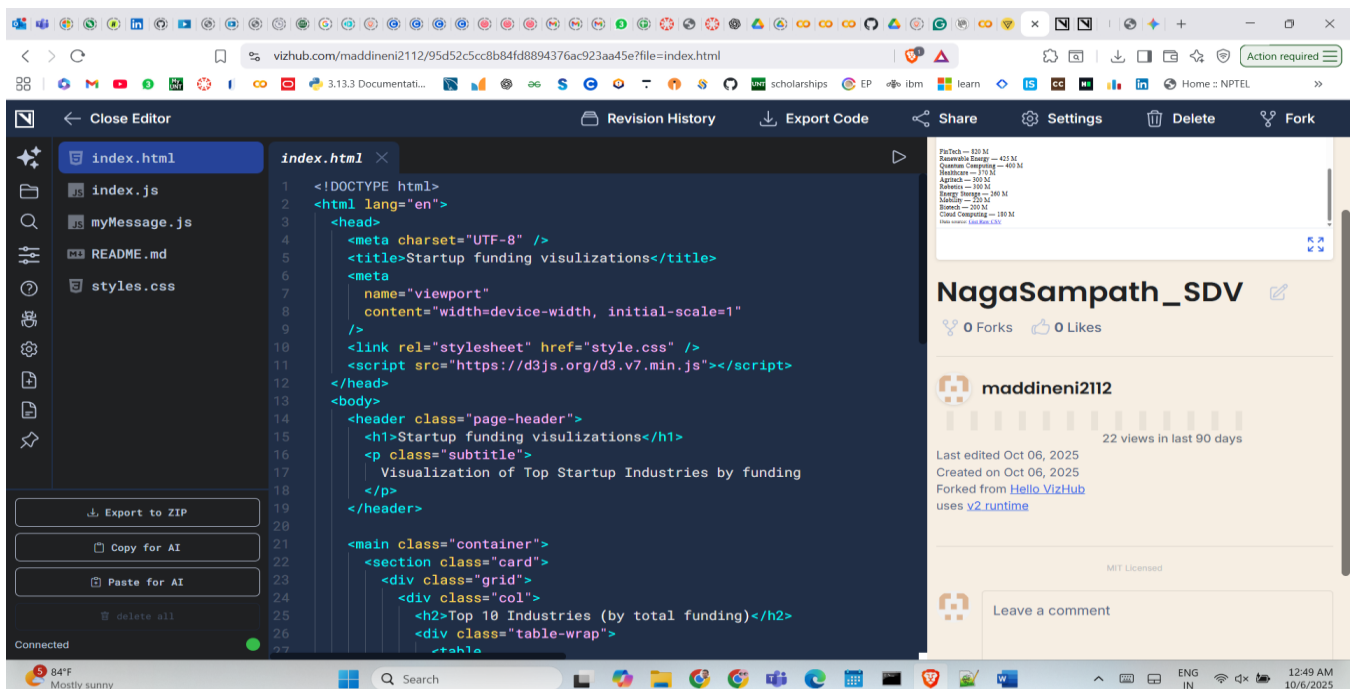


The screenshot shows the VizHub website. The main heading is 'Visualize Data with Generative AI Power'. Below it, there's a subheading 'Transform Ideas into Visualizations'. There are three buttons: 'Instant Results', 'Edit with AI', and 'Revision History'. At the bottom, there are two buttons: 'Create visualization' and 'Learn More'. On the right side, there is a video player showing a visualization of 'Elegant Purple Circles' with a 'Copy link' button. The video title is 'Fork of Circles with D3'.

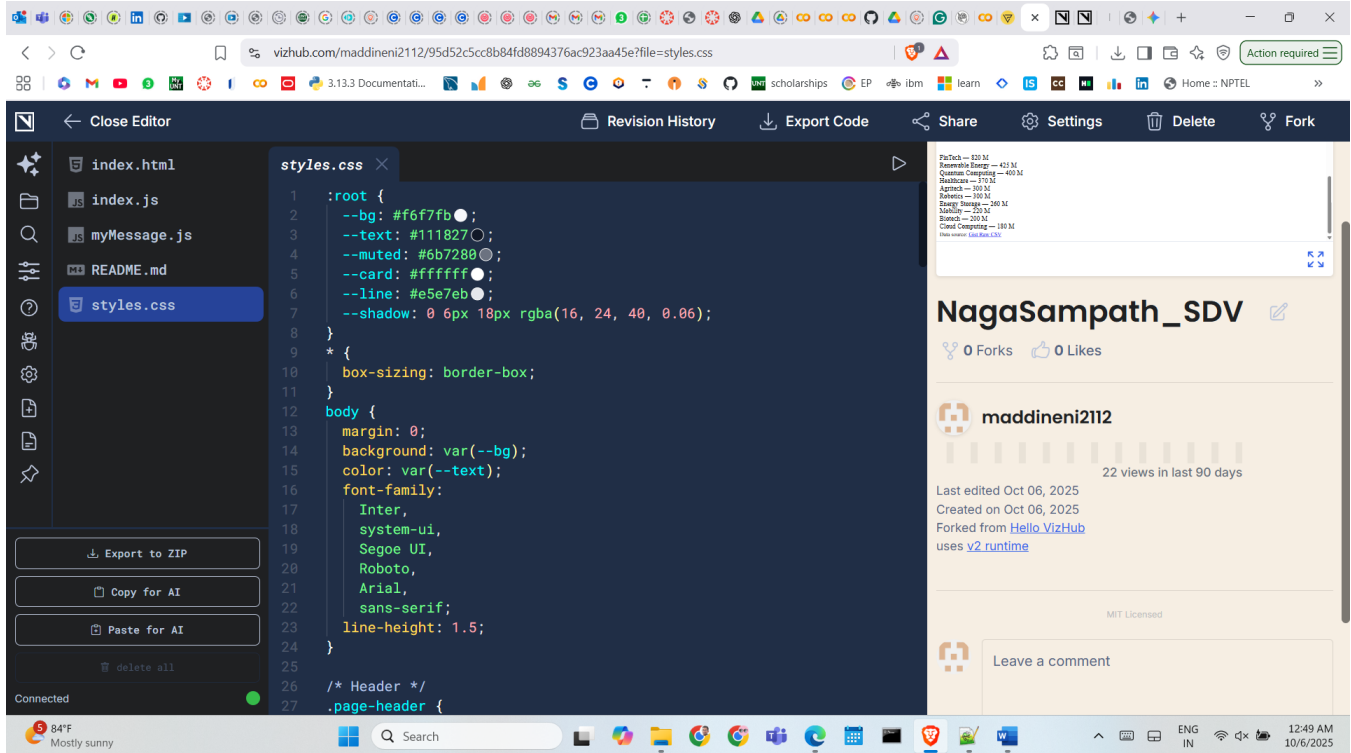
- Next a select a basic template and fork it, you should be able to find the 3 files in it index.html, style.css, index.js files



- Now in the index file we can paste our HTML code as shown below. You be able to see the heading that you created.

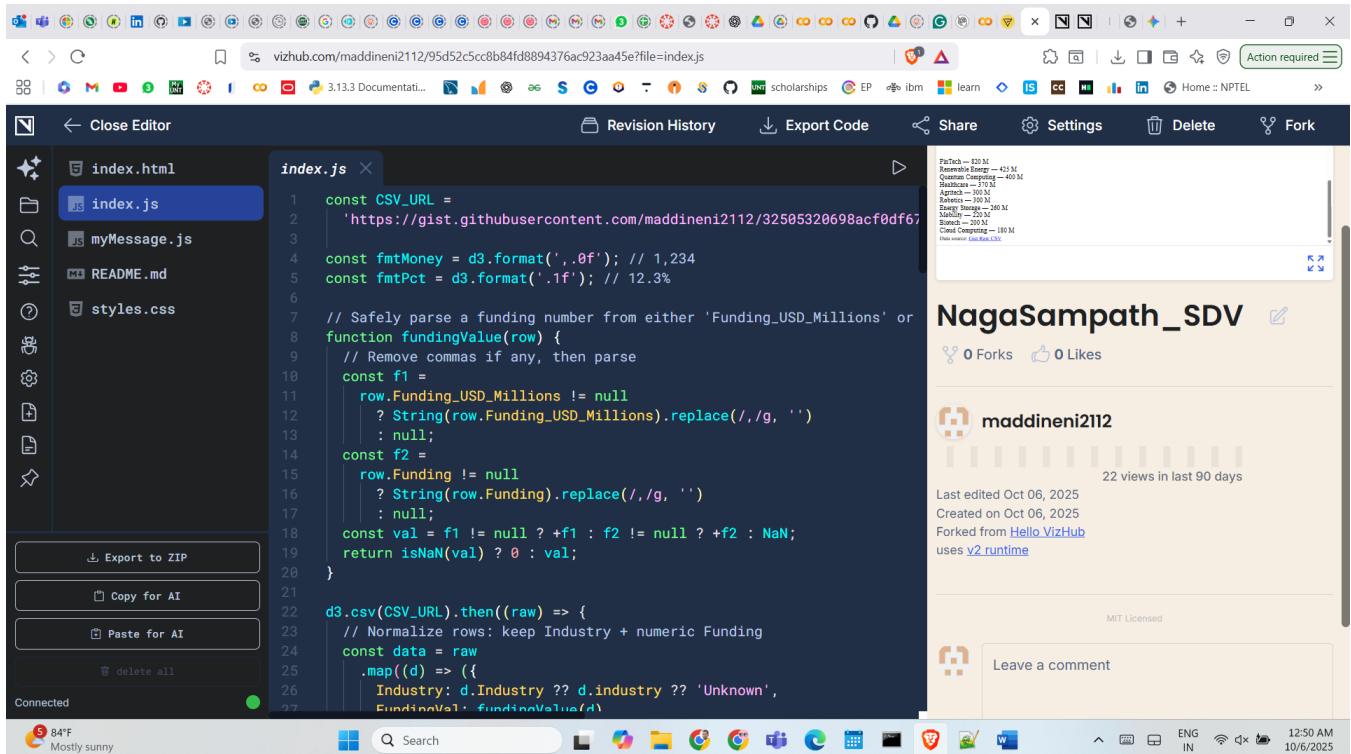


6. Similarly we get the CSS code. In style.css file



```
1 :root {
2   --bg: #f6f7fb;
3   --text: #111827;
4   --muted: #6b7280;
5   --card: #ffffff;
6   --line: #e5e7eb;
7   --shadow: 0 6px 18px rgba(16, 24, 40, 0.06);
8 }
9 * {
10  box-sizing: border-box;
11 }
12 body {
13  margin: 0;
14  background: var(--bg);
15  color: var(--text);
16  font-family:
17    Inter,
18    system-ui,
19    Segoe UI,
20    Roboto,
21    Arial,
22    sans-serif;
23  line-height: 1.5;
24 }
25
26 /* Header */
27 .page-header {
```

7. In js file we do the code and embed svg of a table and pie chart into our page as the tutorial criteria



```
1 const CSV_URL =
2   'https://gist.githubusercontent.com/maddineni2112/32505320698acf0df6';
3
4 const fmtMoney = d3.format(',.0f'); // 1,234
5 const fmtPct = d3.format('%.1f'); // 12.3%
6
7 // Safely parse a funding number from either 'Funding_USD_Millions' or
8 function fundingValue(row) {
9   // Remove commas if any, then parse
10  const f1 =
11    row.Funding_USD_Millions != null
12    ? String(row.Funding_USD_Millions).replace(/,/g, '')
13    : null;
14  const f2 =
15    row.Funding != null
16    ? String(row.Funding).replace(/,/g, '')
17    : null;
18  const val = f1 != null ? +f1 : f2 != null ? +f2 : NaN;
19  return isNaN(val) ? 0 : val;
20 }
21
22 d3.csv(CSV_URL).then((raw) => {
23   // Normalize rows: keep Industry + numeric Funding
24   const data = raw
25     .map((d) => ({
26       Industry: d.Industry ?? d.industry ?? 'Unknown',
27       FundingVal: fundingValue(d)
```

Startup funding visualizations

Visualization of Top Startup Industries by funding

Top 10 Industries (by total funding)

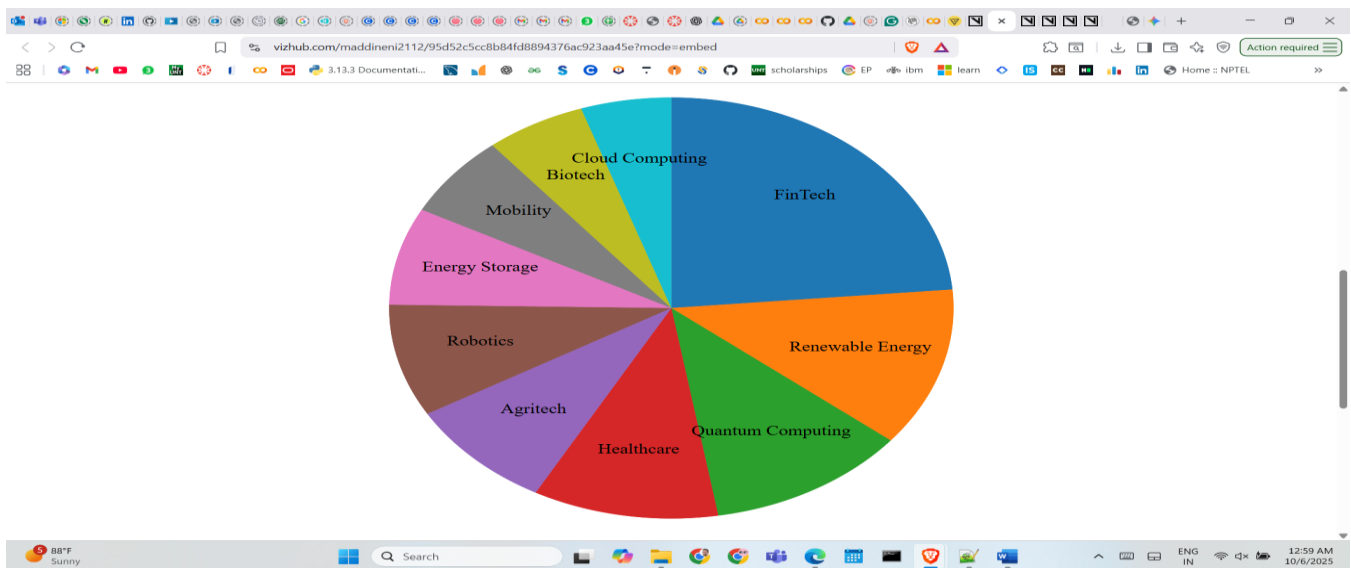
#	Industry	Total Funding (USD M)
1	FinTech	820 M
2	Renewable Energy	425 M
3	Quantum Computing	400 M
4	Healthcare	370 M
5	Agritech	300 M
6	Robotics	300 M
7	Energy Storage	260 M
8	Mobility	220 M
9	Biotech	200 M
10	Cloud Computing	180 M

Share of Top 10 Industries (by funding)

Data Analysis:

The dataset highlights the distribution of total funding across different startup industries. When aggregated, a small set of industries dominate the funding landscape, with the top ten sectors capturing the vast majority of investment capital. These leading categories, which include technology-focused fields like software, e-commerce, and fintech, show disproportionately high funding compared to other industries, suggesting that investor interest is concentrated in innovation-driven areas with high growth potential. The table ranking these top 10 industries by funding gives a clear numeric perspective, making it easy to see which sectors attract the largest sums.

The pie chart complements the table by visually emphasizing each industry's share of the overall top-10 funding pool. Large slices highlight dominant sectors, like fintech being the highest followed by renewable energy, quantum computing and Healthcare industries while smaller slices represent those receiving comparatively modest allocations biotech, mobility etc. This visual makes the imbalance in funding distribution more intuitive, showing briefly which industries have emerged as primary recipients of venture capital. Together, the data and chart reveal not only the scale of investment but also the strategic focus of the startup ecosystem, where resources cluster around technology-intensive and consumer-driven markets



Viz hub link: <https://vizhub.com/maddineni2112/95d52c5cc8b84fd8894376ac923aa45e>